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"What does finding a cure for triple-negative breast cancer look like? It looks like research. That's why I do what I do. "

Sharon Rivera-Sanchez never dreamed she'd become a voice for diversity in medical research. But six years after her breast cancer diagnosis, the former finance professional has found her calling:

Encouraging women of color to take part in clinical trials.

It started with a routine mammogram

Doctors had discovered a benign cyst during Sharon Rivera-Sanchez's annual mammogram in 2013. Still, Sharon had no reason to think her preventative mammogram two years later would find anything concerning.

Unfortunately, the scan proved otherwise. A mass on Sharon's breast turned out to be triple-negative breast cancer, a rare and aggressive cancer with high recurrence rates. Sharon was only in her 50s, which is younger than the median age for women diagnosed with breast cancer.

The wife and mother of four adult children says her entire life "went on hold" in an instant.

"My focus became, 'What does a fight look like?' You always hear people say you've got to 'fight' cancer, but what does that really mean?"

"I didn't want to learn from the internet"

For Sharon, it meant research.

She scheduled surgery to have her tumor removed. Before she'd pursue the recommended chemotherapy, though, she needed to learn more. Years prior, she'd seen her stepfather endure debilitating side effects from chemotherapy. Sharon, who lives in Virginia, began educating herself on her options – *away* from her keyboard.

"I didn't want to learn from the internet," she says. "I signed up for seminars. I scheduled consultations at top healthcare systems in New York City and Baltimore. I went back to my surgeon twice with a list of questions."

Triple-negative breast cancer's name nods to the fact that its cells lack the three receptors – estrogen, progesterone, and a protein called HER2 – found in most other breast cancer types. Without the ability to target these receptors, there are few treatment options other than chemotherapy. Ultimately, it was an oncologist in Baltimore who convinced Sharon that chemotherapy was, indeed, her best course of action.

"She looked at me and said, 'If anyone says you don't need chemotherapy, they're lying.' I could tell she really cared."

After the tumor in her breast was removed, Sharon began receiving chemotherapy at a hospital close to her home outside of Richmond, where she'd be near her doctors in the event of complications or side effects. There *were* side effects – namely, neuropathy in her hands and feet.

"It got so severe that I couldn't undergo my final two rounds of chemo," Sharon remembers. "Even though I was in remission at that point, that concerned me."

Throughout chemotherapy, Sharon had been asking doctors about something she'd discovered in her research: breast cancer clinical trials. Clinical trials are research studies designed to test the effectiveness of breast cancer treatments that look promising in the lab, but have yet to be tested on humans. Participants sign up in hopes that the experimental intervention will not only improve their own health outcome, but will work for other patients and become an approved treatment option for all.

When Sharon asked if she might qualify for any trials, she kept getting the same answer: there were only a few studies for triple-negative breast cancer. That she was better off finishing chemotherapy, getting back to regular life, and hoping the cancer didn't return.

Which she tried to do – unsuccessfully.

"Triple-negative breast cancer has an extremely high recurrence rate," Sharon says. "I couldn't just sit back and wait for it to return. I was like, 'What's next?'"

Then, watching TV one day, Sharon caught an interview with renowned oncologist Dr. Kristi Funk. Dr. Funk was discussing a new clinical trial for triple-negative patients and survivors. The trial's focus was a blood test

designed to detect signs of recurrence, allowing doctors to more actively monitor patients whose cancer had a high likelihood of returning.

"I was like, 'Oh my God, there is something proactive I can do!"

There was just one problem: The trial was in California.

Hope – and a hurdle

Sharon was accepted into the clinical trial. It meant traveling from Virginia to California for two to three days every 90 days – a costly undertaking. After two trips funded largely by a GoFundMe set up by her children, Sharon realized it was too cost-prohibitive to continue.

"I didn't know what to do. I didn't want to ask people for the money I needed to travel to California more than a dozen more times." Another California trial participant, a woman from Pennsylvania named Faith, also dropped out because of travel costs. It was Faith who first mentioned a name that would soon play a pivotal role in Sharon's health journey:

Dr. DeMichele.

"I said, 'Who's that?'" Sharon remembers. "And Faith told me, 'She's got a study at Penn Medicine in Philadelphia.'"

Meanwhile, in Pennsylvania

While Sharon was navigating treatment and research options, something groundbreaking was happening at Penn Medicine's Abramson Cancer Center: Nationally renowned cancer researchers Angela M. DeMichele, MD, MCSE and Lewis A. Chodish, MD, PhD were establishing the **2-PREVENT T**ranslational **C**enter of **E**xcellence, called 2-PREVENT for short. 2-PREVENT's mission is to use new technology and novel therapeutic trials – from prediction and prevention to treatment and monitoring – to understand why some women, but not others, experience breast cancer recurrence.

"Currently, women with triple-negative breast cancer have a high risk of recurrent, metastatic disease after they are initially treated. This trial utilizes a test for dormant or 'sleeping' cells in the bone marrow to identify women at risk. Our goal is to kill these cells and harness the immune system to eliminate the cancer once and for all," Dr. DeMichele explains.

The formal name for these sleeping cells is Disseminated Tumor Cells (DTC for short). When present in bone marrow, DTCs often defy chemotherapy and other primary cancer therapies, then remain in a dormant state

indefinitely. Their presence puts a patient at increased risk for cancer relapse.

One of 2-PREVENT's ongoing therapeutic trials – the CLEVER study – is a Phase II trial looking into the ability of two drugs (HydroxyChLoroquine and EVErolimus) to prevent breast cancer recurrences in patients with DTC cells by targeting and hopefully destroying them. It's the trial Faith was referring to when she told Sharon about Dr. DeMichele.

The call that changed it all

Sharon picked up the phone and dialed the intake line at Penn Medicine. She received information about the CLEVER study, answered initial screening questions, and was told the next step was an evaluation to determine if there were, in fact, DTC "sleeping cancer cells" in her bone marrow.

Sharon and her husband drove from Virginia to her appointment at Penn Medicine on April 12, 2017. Three weeks after her bone marrow draw, she received the results. Sharon had three dormant DTC cells, which indicated a heightened risk of cancer returning. The silver lining: She qualified for the CLEVER clinical trial, which would aim to destroy them. As part of the trial, Sharon would take pills for six to 12 months. This could begin as soon as she was properly educated on the process and risks, and once she gave her consent.

"One thing I always say about Penn Medicine: They made it clear to me from the very beginning that this isn't a miracle or even a treatment. This is a *trial*. We are trying to *find* new and effective treatments. I remember joking with Dr. DeMichele, 'Are you trying to talk me out of this?' because there were so many amendments and consent forms I had to sign in front of her and a nurse witness, showing I agreed to all of the risks and possible outcomes. But I appreciated that."

Sharon was enrolled in the CLEVER study on May 30, 2017.

The trial was randomized, meaning participants were randomly chosen to receive one of four treatment arms, or plans. Sharon's study arm received both drugs – HydroxychLoroquine and Everolimus – daily for 24 weeks (about six months). She'd take the orally-administered pills at home in Virginia and travel to Philadelphia every four weeks for monitoring.

After 24 weeks, another bone marrow draw revealed exciting news: Two of the three dormant DTC cells had been destroyed by the experimental treatment.

She'd continue the medication and monitoring for another six months.

Looking around

It was around this time that Sharon noticed something about the CLEVER study: There was only one other Black woman among the participants.

"That's when I started looking at data related to race," she remembers. "I discovered that the majority of cancer trial participants are white women."

This wasn't a surprise to the Abramson Cancer Center (ACC), who's been proactively working for five years to increase the number of Black participants in clinical trials. Those efforts have begun to pay off. The number of Black cancer treatment trial participants at the ACC has almost doubled, from 12.2 percent a few years ago to

20.9 percent today. Nationally, though, the number of Black clinical trial participants still lags far behind, at just 4 percent to 5 percent. It's a gap believed to be attributed in part to distrust of the healthcare system among some Black Americans.

Sharon also noticed another reality facing fellow cancer patients: associated costs.

"Cancer is an expensive disease to have," she says. "Beyond co-payments and things like childcare and transportation and time off of work, there are specific brands of personal care items like mouthwash and lotion that are recommended during treatment. But buying them so you can try them out and find the ones that works for you can get very expensive."

Sharon began cutting coupons and buying these items in bulk. She'd go down the line in the waiting room, handing them out to other patients and trial participants.

A Penn Medicine nurse practitioner named Brooke took notice.

"Brooke said to me, 'You don't have to go into your own pocket for that. Why don't you start a nonprofit?'"

One step closer to her calling

Sharon started Saving Pennies 4 a Cure, a 501(c)(3) that gives household and personal care items to patients actively receiving chemotherapy.

"But I still kept finding myself being pulled toward the clinical trial and research side of things," she remembers.

Lack of diversity in clinical trials especially bothered her. She knew her own life experience could help alleviate it.

"Today I live in an upscale area, but I haven't always. I've lived in the projects. I have seen firsthand the lack of trust that the Black community has about medical research in particular, and at some level, I get it."

When she told them she was taking part in a clinical trial, some of Sharon's friends and acquaintances brought up the infamous Tuskegee Experiment (a study in which Black men were under the impression they were being treated for syphilis, but in actuality, were not) and Henrietta Lacks (a Black woman whose cells, unbeknownst to her, were extracted and used in widespread medical research).

"They'd say things like, 'You could never make me a guinea pig like that,'" Sharon says. "I'd remind them that those examples were a long time ago. Laws are in place now; there's the Food and Drug Administration and the National Cancer Institute, which has Black people like me involved. I'd also ask, 'Do you take aspirin?' When they said yes, I'd remind them that we only have aspirin because it went through clinical trials."

Sharon started a second non-profit: Trials of Color, aimed at bridging the gap between clinical research and people of color through advocacy, education, and giving back to the cancer community.

Sharon and her team of volunteers visit infusion and chemotherapy rooms at various healthcare systems and initiate conversations with minority patients – some of whom, she says, have never heard of clinical trials. Others say they can't consider a trial because they have to work, or don't have transportation or childcare. Sharon takes steps to connect those patients with resources that can help.

"There are so many types of clinical trials out there, testing everything from exercise to prevention. I tell them, 'Go back and ask your doctor about trial options.'"

Most of all, she tries to alleviate larger distrust.

"Research is so important, and it's only going to be accurate if people of color take part in it."

Leading researchers like Dr. DeMichele are grateful.

"Sharon's work is so important to our research effort at the ACC. It is vital that women from different backgrounds participate in research. Otherwise we won't know if the results of trials that lead to new treatments apply to them. Sharon's willingness to share her experience and knowledge is making such a difference" says Dr. DeMichele.

CLEVER wraps up – but Sharon's work is only beginning

For the remaining six months of the CLEVER study treatment at Penn Medicine, Sharon and her husband drove or took the train from Virginia to Philadelphia for her appointments. She experienced one temporary side effect from the trial drugs – flashing lights in her vision – but otherwise felt good throughout. Sharon's last trial visit was in August of 2021; she'll visit the Abramson Cancer Center for follow-up appointments every six months for three years.

At the end of the CLEVER study, Sharon had a choice: She could find out whether the one dormant cell remained in her bone marrow, or if it had been destroyed by the intervention. She opted not to know.

Most importantly, Sharon's cancer is still in remission – which she attributes to her participation in the CLEVER study. Her fondness for her trial team is just the icing on the cake.

"I love the Abramson Cancer Center – especially Dr. DeMichele and her team: Brooke, Isoris, Nancy, and Jessica. I feel like we're one huge clinical trials family. From the moment you pull up to Penn Medicine's valet parking to the beautiful, live piano music in the lobby…I just get a beautiful, warm feeling. You are truly treated like a partner in research – not a number – at Penn."

Today Sharon is nothing short of busy. She serves on three boards: The Massey Cancer Center NCORP board, the National Cancer Institute's Prevention Board, and the board for the RadComp Trial Study – short for Radiotherapy Comparative Effectiveness – which seeks to learn whether proton or photon radiation therapy will be most beneficial to patients' quality of life and long-term health. She is a Grasp Cancer mentor and is also in training for the Clinical Trial Ambassador Program at Penn Medicine, which seeks to eliminate racial inequity in clinical trials.

Ultimately, Sharon would like to take part in another trial herself.

"I want Dr. DeMichele to find me another one!" she says with a laugh. "I love being monitored because we're talking about a life-threatening disease with a high rate of recurrence. Plus, we all learn and benefit from clinical trials."

She says the best advice she can give to women of color who have been diagnosed with cancer is to get involved in research.

"Don't just sit and wait," she says. "At some level research is a crap shoot, yes, but when you get involved, you're making a difference and potentially improving your outcome at the same time."

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